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Dr. Kaitlin McCardle
Senior Editor
Nature Computational Science
Springer Nature, United States of America

RE: REVISION – NATCOMPUTSCI-26-0870A, "HIP: Hessian Interatomic Potentials without derivatives"

Dear Dr. McCardle and Reviewers,

On behalf of my co-authors, I am pleased to submit our revised manuscript. We thank the reviewers for their positive assessments and Reviewer 1 for the constructive suggestion and careful corrections.

In response, we added a Discussion paragraph proposing approximate Hessian pseudo-label pre-training followed by fine-tuning on a smaller set of reference DFT Hessians. We also resolved the figure and table cross-references and standardized the capitalization of "Hessian" throughout the manuscript and Supplementary Information.

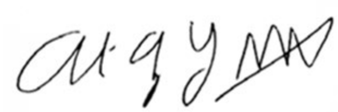
The requested summary for the journal website and e-alerts is: "Researchers report HIP, a machine-learning model that predicts molecular Hessians directly rather than by differentiation. HIP is faster and more accurate for geometry optimization, transition-state search and vibrational analysis."

For social-media promotion, the first author can be tagged on X (@atAndreasBurger) and Bluesky (@andreasburger.bsky.social); the Matter Lab can be tagged on Bluesky (@thematterlab.bsky.social). Suggested hashtags are #ComputationalChemistry, #MachineLearning and #AIforScience.

We provide a point-by-point response (in blue) below.

Thank you for your time and consideration.

Sincerely,
Alán Aspuru-Guzik
on behalf of all authors



Response to reviewers

Reviewer 1: Remarks to the Author

I thank the authors for their careful and thorough revision. They have addressed all of the points raised by the other reviewer. I am happy to recommend it for publication.

We thank Reviewer 1 for the careful assessment and recommendation for publication.

Suggestion It could be worth exploring whether a Hessian training set can be bootstrapped from an accurate model rather than relying entirely on ground-truth DFT Hessians. Concretely, one could take an energy-and-force model that is not trained on Hessians but already produces qualitatively reasonable Hessians via automatic differentiation (for instance one of the AD models in Table 1), use it to generate an initial set of approximate Hessian labels, and then fine-tune HIP on a small amount of ground-truth DFT Hessian data. The interesting question is whether such a pre-training/fine-tuning scheme could reach the same Hessian accuracy as training on a large ground-truth dataset, but with far fewer expensive DFT Hessian calculations. Given the authors' central argument that Hessian labels are the practical bottleneck, this would be a valuable direction for reducing the data cost, and even a brief comment or small proof-of-concept would be a welcome addition.

We thank the reviewer for this valuable suggestion. We agree that pseudo-label pre-training could reduce the number of reference DFT Hessians required. We added a paragraph to the Discussion proposing automatic-differentiation Hessians from an energy-and-force model as approximate pre-training labels, followed by fine-tuning on a smaller set of reference DFT Hessians. We also note that the value of this strategy will depend on whether fine-tuning can correct curvature errors in the AD Hessians. A controlled study of pseudo-label quality and final Hessian accuracy is therefore an important direction for future work.

Minor correction 1 There are several broken cross-references in the manuscript, appearing as "Fig. ??" and "Table ??" (for example in the geometry optimization, transition state search, and zero-point energy sections). These should be resolved before publication.

We corrected the cross-references.

Minor correction 2 "Hessian" is inconsistently capitalised throughout the text and is in several places written in lower case ("hessian"). This should be made consistent.

We standardized "Hessian" and "Hessians" to initial capitals throughout the manuscript, Supplementary Information and reference titles.

I congratulate the authors on a strong contribution.

We thank the reviewer for the encouraging assessment.

Reviewer 2

No comments were provided.

Reviewer 3: Remarks to the Author

My comments have all been addressed, appreciate that the authors have made the modifications to make it more accessible!

We thank Reviewer 3 for the positive assessment and for confirming that the previous comments have been addressed.

Reviewer 3: Remarks on code availability

Code installs more cleanly now, thanks for updating the procedure.

We thank the reviewer for testing the updated installation procedure.